

1. Administrative & Study Identification

Full Study Title: A Clinical Trial to Learn About the Effects of VHB937 in People With Amyotrophic Lateral Sclerosis (ALS) **Short Title/Acronym:** ASTRALS Trial **Protocol Number:** CVHB937B12201 **NCT Number:** NCT06643481 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** VHB937 (TREM2-targeted monoclonal antibody) **Phase of Development:** Phase II **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of VHB937 compared to placebo in participants with early-stage Amyotrophic Lateral Sclerosis (ALS). **Secondary Objectives:** To assess the safety, tolerability, and effects on functional decline, survival, and respiratory capability.

2.2 Background & Rationale To address the urgent need for new therapies in early-stage Amyotrophic Lateral Sclerosis (ALS). VHB937 provides a unique approach by acting as a monoclonal antibody that targets and stabilizes TREM2 on microglia. This potentially targets underlying neuroinflammatory pathways to slow disease progression, unlike traditional medications that may only manage symptoms.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized (Parallel group)

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 225-251 participants **Number of Sites:** Multicenter (Global/Regional) **Estimated Study Start:** Late 2024 **Estimated Primary Completion:** 40-week double-blind treatment period

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years or older. 2. Confirmed diagnosis of ALS with symptoms (weakness) present for ≤ 24 months prior to trial entry. 3. Mild symptoms of ALS as measured by an ALSFRS-R questionnaire total score ≥ 30 . 4. Breathing function/vital capacity $\geq 60\%$ of what is expected for the participant's sex, height, and age.

4.2 Exclusion Criteria 1. Use of other investigational drugs within 5 half-lives of screening or 30 days. 2. Women of child-bearing potential not using highly effective contraception. 3. History or current diagnosis of significant cardiac conditions, clinical evidence of liver or renal disease/injury, or severe respiratory disease (e.g., COPD, pulmonary fibrosis). 4. Presence of unstable psychiatric disease, cognitive impairment, or suicidal ideation.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Administered via infusion every 4 weeks. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** 40-week double-blind period, followed by an open-label extension (OLE).

5.2 Concomitant & Prohibited Therapy Permitted: Stable dose of an approved treatment for ALS. **Prohibited:** Other investigational drugs and defined prohibited medications.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to 4 weeks	Informed Consent, Medical History, Breathing tests, Lumbar Puncture
Baseline	Day 0	Randomization, First Infusion, Baseline ALSFRS-R
Interim	Every 4 weeks	Infusion dosing, Safety Labs, Efficacy Assessment, Vital Signs, Hand grip strength
End of Study	Week 40	Final Double-Blind Assessments, Option to continue to Open-Label Extension

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Efficacy evaluated by a composite of permanent assisted ventilation-free survival up to week 40 and change in ALS Functional Rating Scale-Revised (ALSFRS-R). **Measurement Method:** Clinical assessment and ALSFRS-R questionnaire scoring.

7.2 Secondary & Exploratory Endpoints Change in vital capacity / breathing tests. Biomarker changes (e.g., sTREM2 and neuroinflammatory markers). Incidence and severity of Adverse Events (Safety and Tolerability).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Regular monitoring via physical & neurological exams, vital signs, ECG, and laboratory tests

(blood/urine). **Serious Adverse Events (SAEs):** Standard reporting timelines as per Novartis/GCP protocols (typically 24 hours).
Data Monitoring Committee (DMC): Independent safety review committee to monitor ongoing safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy, Safety Set for adverse event monitoring. **Sample Size Justification:** Appropriately powered to detect meaningful shifts in ALSFRS-R scores and survival outcomes over 40 weeks with ~225-251 patients. **Handling of Missing Data:** Standard handling via Mixed-Model Repeated Measures (MMRM) or Multiple Imputation for dropouts.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
Monitoring Plan: Onsite monitoring coordinated with the 4-week infusion schedules. **Audit Trail:** eCRF locking and query resolution via standard sponsor data management protocols.

11. Study Identifiers & Governance

Primary ID: CVHB937B12201 **Registry Number:** NCT06643481 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** ASTRALS Trial **Official Regulatory Title:** A Clinical Trial to Learn About the Effects of VHB937 in People With Amyotrophic Lateral Sclerosis (ALS)

12. Clinical Framework (ALS Specific)

Primary Condition: Amyotrophic Lateral Sclerosis (ALS) **Diagnosis Threshold:** ALSFRS-R ≥ 30 , symptom onset ≤ 24 months
Medical Methodology: TREM2 stabilization and activation therapy / Standard ALS Care **Optimization Target:** Slowing disease progression / preserving functional capacity

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for ALS + Investigational infusions **Education Protocol (HCP):** Site training for IV administration and neurological assessments **Education Protocol (Patient):** Informed consent, treatment expectations, and trial procedures education **Quality Audit Mechanism:** Ongoing clinical monitoring and safety reviews

Role	Name	Signature	Date	:	---	:	---	:	---	:	---
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]	:	---	:	---	:	---	:	---
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]	:	---	:	---	:	---	:	---
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]	:	---	:	---	:	---	:	---